A Robust Linear Upper Bound for the Variance of Random k-SAT Hitting Times

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Abstract

Fix an integer $k \geq 3$. For each n, consider an independent, not necessarily identically distributed stream of proper signed k-clauses on n Boolean variables. Suppose that every clause law is bounded above and below, pointwise, by fixed constant multiples of the uniform proper-clause law. We prove that the first unsatisfiable prefix H_n satisfies

$$\mathrm{Var}(H_n) = O(n),$$

uniformly over all such streams. The same bound holds for H_n clipped at any fixed linear time horizon. The argument robustifies the clause-omission and root-test method developed for uniform random k-SAT and the killing potential introduced for the sharp three-literal variance bound. The main new ingredients are a general-k potential inequality with exponent $k/(k-1)$ and a conditional-density analysis showing that nonuniform root tests contribute only an exponential factor in a root-incidence count with uniformly bounded exponential moments. We also isolate a strictly more local robustness criterion: global pointwise upper comparison can be replaced by conditional test-density and root-incidence bounds. No stationarity in the clause laws is required.

1 Introduction

Let $x_1, \ldots, x_n$ be Boolean variables. A *proper signed k-clause* is a disjunction of k literals on k distinct variables. In the classical random k-SAT process, independent clauses are sampled uniformly with replacement, and one studies the first prefix whose conjunction is unsatisfiable. Sharp threshold phenomena for random k-SAT originate in the work of Friedgut [1999]; quantitative lower bounds on the width of the uniform transition were obtained by Wilson [2002]. Recent work of Carenini [2026b] and Carenini [2026a] gives substantially sharper scaling-window bounds and establishes the fixed-k satisfiability conjecture.

The fluctuation problem for the first unsatisfiable prefix is finer than the existence or width of a threshold. In the uniform proper-clause model, OpenAI [2026a] proved variance $\Theta_k(n)$ for fixed $k \geq 4$ and an $O(n \log n)$ upper bound for $k = 3$. The logarithmic loss in the three-literal case was subsequently removed in OpenAI [2026b], yielding variance $\Theta(n)$. Their upper-bound arguments combine coordinate omission, necessary tests obtained by flipping variables of a deleted clause, and an occupation estimate for a potential defined on the surviving assignment set.

The purpose of this paper is to show that the linear upper bound is stable under bounded, time-dependent perturbations of the complete clause law. The clause stream may be nonstationary, its variable and sign choices may exhibit arbitrary dependence within a clause, and different times

may use different distributions. Independence *between* clause times is retained. The only structural assumption is a two-sided pointwise density bound relative to the uniform proper-clause law.

1.1 Model and main result

Let $\mathcal{C}_{n,k}$ denote the set of proper signed k-clauses on $x_1, \dots, x_n$, and let $U_{n,k}$ be the uniform probability measure on $\mathcal{C}_{n,k}$:

$$U_{n,k}(C) = \frac{1}{2^k \binom{n}{k}} \qquad (C \in \mathcal{C}_{n,k}).$$

For every n, let $C_1, C_2, \dots$ be independent clauses, where C_j has law $\mu_{n,j}$ on $\mathcal{C}_{n,k}$. We assume that fixed constants

$$0 < a \le 1 \le b < \infty$$

satisfy

$$a\, U_{n,k}(C) \le \mu_{n,j}(C) \le b\, U_{n,k}(C) \tag{1}$$

for every n, every time j, and every $C \in \mathcal{C}_{n,k}$. Put

$$F_m = \bigwedge_{j=1}^{m} C_j, \qquad H_n = \min\{m \ge 1 : F_m \text{ is unsatisfiable}\}.$$

Theorem 1.1 (Robust linear variance bound)**.** *Fix $k \ge 3$ and constants $0 < a \le 1 \le b < \infty$. Under (1), there is a constant $K = K(a, b, k) < \infty$ such that*

$$\operatorname{Var}(H_n) \le Kn \qquad (n \ge k).$$

More generally, for every fixed $\rho < \infty$, there is a constant $K_\rho = K_\rho(a, b, k) < \infty$ such that

$$\operatorname{Var}(H_n \wedge L) \le K_\rho n$$

for all $n \ge k$ and all integers $1 \le L \le \rho n$.

The constants are uniform over the entire triangular array $(\mu_{n,j})_{n,j}$. In particular, deterministic time dependence is allowed without any regularity assumption in j.

The proof yields a more local theorem. For a k-set R of roots, reveal from a clause the subset of R that it uses and the literal signs on that subset. Conditional on this revealed root trace, the remaining literals are called the hidden nonroot part.

Theorem 1.2 (Local robustness criterion)**.** *Fix $k \ge 3$. Suppose the clauses are independent across time and that there are constants*

$$B_{\mathrm{o}}, B_{\mathrm{t}} \ge 1, \qquad \gamma_1, \gamma_2, \delta > 0$$

with the following properties, uniformly in n, the time index, every k-set $R \subseteq [n]$, and every relevant root trace of positive probability:

(i) *conditional on avoiding R, the hidden proper k-clause on the $n - k$ nonroots has density at least B_{o}^{-1} relative to the uniform law;*

(ii) *conditional on using exactly one specified root with a specified sign, the hidden proper $(k-1)$-clause has density at most B_{t} relative to the uniform law;*

(iii) *a clause meets R with probability at most γ_1/n, and meets at least two elements of R with probability at most $\gamma_2/[n(n-1)]$;*

(iv) *for every fixed assignment and every time, the sampled clause falsifies that assignment with probability at least δ.*

Then, for every fixed $\rho < \infty$,

$$\mathrm{Var}(H_n \wedge L) = O(n) \qquad (1 \le L \le \rho n),$$

where the constant depends only on $k, \rho, B_{\mathrm{o}}, B_{\mathrm{t}}, \gamma_1, \gamma_2$. Moreover $\mathrm{Var}(H_n) = O(n)$, with dependence also on δ.

Under (1), the theorem applies with

$$B_{\mathrm{o}} = B_{\mathrm{t}} = \frac{b}{a}, \qquad \gamma_1 = bk^2, \qquad \gamma_2 = \frac{bk^2(k-1)^2}{2}, \qquad \delta = a2^{-k}. \tag{2}$$

Indeed, conditional densities lie in $[a/b, b/a]$; the incidence estimates follow by union bounds over roots and root pairs; and a fixed assignment falsifies a uniform proper clause with probability 2^{-k}. Thus Theorem 1.1 is a direct corollary of the local criterion.

The variance estimate immediately controls every fixed central quantile window. Define

$$s_n(m) = \mathbb{P}(H_n > m), \qquad r_n(p) = \min\{m \ge 0 : s_n(m) \le p\}, \quad 0 < p < 1.$$

Corollary 1.3 (Central windows)**.** *Under the hypotheses of Theorem 1.1, for every fixed $0 < \varepsilon < 1/2$,*

$$r_n(\varepsilon) - r_n(1-\varepsilon) = O_{a,b,k,\varepsilon}(\sqrt{n}).$$

We prove only an upper variance bound. The linear lower bound known for the uniform model uses additional information about separation of satisfiability levels and does not automatically persist for every nonidentical stream satisfying (1). Although the lower density bound permits the mixture representation

$$\mu_{n,j} = aU_{n,k} + (1-a)\nu_{n,j},$$

Bernoulli thinning alone does not give a variance lower bound: the residual clauses may determine the hitting time before fluctuations of the uniform component become pivotal. Establishing a matching lower bound therefore requires a separate anti-concentration or level-separation argument and remains open in this generality.

1.2 Proof mechanism

Fix one deleted clause and call its k variables *roots*. If that clause is pivotal, then every surviving solution has the root assignment that falsifies it. Flipping an available root produces a necessary family of $(k-1)$-clause tests on the nonroot solution set. If p denotes the probability that d independent uniform $(k-1)$-clauses kill that set, the p-dependence of the argument is

$$\underbrace{p^{k}}_{\text{necessary root tests}} \times \underbrace{p^{-k/(k-1)}}_{\text{remaining duration}} \le \underbrace{p^{k/(k-1)}}_{\text{controlled occupation}}.$$

The inequality is exactly the exponent condition

$$k - \frac{k}{k-1} \ge \frac{k}{k-1},$$

which holds for every $k \geq 3$ and is an equality at $k = 3$.

The local hypotheses have distinct roles. The lower conditional density B_{o}^{-1} drives the occupation potential at ordinary clauses. The upper conditional test density B_{t} costs a factor at most B_{t}^{r} for a family of r hidden root tests. The incidence and collision bounds make the total root-incidence count exponentially integrable and collisions rare, while δ supplies the tail estimate used to remove the cap. The two-sided pointwise hypothesis is therefore a convenient sufficient condition, not the intrinsic formulation of the argument.

Section 2 proves the general-k killing-potential inequality. Sections 3 and 4 set up the clause deletion and conditional exposure. Sections 5 and 6 prove the root-test and occupation estimates. Section 7 obtains a uniform second-moment bound for each deletion delay, and Section 8 removes the linear cap.

2 A general-k killing potential

We work first on $u \geq k$ Boolean variables, independently of the clause stream. For an assignment set $S \subseteq \{0,1\}^u$ and a clause C, let

$$S[C] = \{z \in S : z \models C\}.$$

For $1 \leq r \leq k$, let $h_r(S)$ be the probability that one uniform proper r-clause kills S, meaning that no member of S satisfies the clause. Set

$$\eta = \frac{k}{k-1}, \qquad c_k = \left(\frac{k-1}{2}\right)^k.$$

Lemma 2.1 (One-clause comparison)**.** *For every nonempty $S \subseteq \{0,1\}^u$,*

$$h_{k-1}(S)^{\eta} \leq k h_k(S) + c_k u^{-k}. \tag{3}$$

Proof. Call a coordinate *frozen* if it has the same value in every assignment of S, and let f be the number of frozen coordinates. A proper r-clause kills all of S exactly when all of its variables are frozen and each literal has the falsifying sign. Therefore

$$h_r(S) = \frac{(f)_r}{2^r (u)_r}, \tag{4}$$

where $(t)_r = t(t-1)\cdots(t-r+1)$, with value zero when the nonnegative integer t is smaller than r.

Since $f \leq u$, each factor $(f-j)/(u-j)$ is at most f/u. Hence

$$h_{k-1}(S)^{1/(k-1)} \leq \frac{f}{2u}. \tag{5}$$

If $f \geq k$, then

$$\frac{h_k(S)}{h_{k-1}(S)} = \frac{f-k+1}{2(u-k+1)} \geq \frac{f}{2ku},$$

because $f - k + 1 \geq f/k$. Combining this with (5) gives

$$h_{k-1}(S)^{\eta} \leq h_{k-1}(S)\frac{f}{2u} \leq k h_k(S).$$

If $f \le k-2$, the left side of (3) vanishes. For the remaining case $f = k-1$, equation (5) gives

$$h_{k-1}(S)^\eta \le \left(\frac{k-1}{2u}\right)^k = c_k u^{-k}.$$

□

For $j \ge 0$, define $q_j(S)$ to be the probability that j independent uniform proper $(k-1)$-clauses kill S. Thus

$$q_j(\varnothing) = 1 \quad (j \ge 0), \qquad q_0(S) = 0 \quad (S \ne \varnothing).$$

The quantity $q_j(S)$ is nondecreasing in j and when S shrinks. For $d \ge 1$, define

$$P_d(S) = \sum_{j=0}^{d-1} q_j(S). \tag{6}$$

Then $0 \le P_d(S) \le d$, and P_d is nondecreasing when S shrinks.

Lemma 2.2 (Uniform potential drift)**.** *Let C be a uniform proper k-clause on the u variables. Then, for every $d \ge 1$,*

$$\mathbf{1}_{\{S \ne \varnothing\}} q_d(S)^\eta \le k d^{\eta-1} \mathbb{E}_C\big[P_d(S[C]) - P_d(S)\big] + c_k d^\eta u^{-k}. \tag{7}$$

Proof. Fix $j \ge 0$, and let Y be the subset of S surviving j independent uniform proper $(k-1)$-clauses. Adding one further such clause gives

$$q_{j+1}(S) - q_j(S) = \mathbb{E}\big[\mathbf{1}_{\{Y \ne \varnothing\}} h_{k-1}(Y)\big]. \tag{8}$$

Adding instead the independent proper k-clause C gives

$$\mathbb{E}_C\big[q_j(S[C]) - q_j(S)\big] = \mathbb{E}\big[\mathbf{1}_{\{Y \ne \varnothing\}} h_k(Y)\big]. \tag{9}$$

Indeed, in either order the increase in killing probability is the probability that the old surviving set is nonempty and the final clause kills it.

By Jensen's inequality and Lemma 2.1,

$$\big(q_{j+1}(S) - q_j(S)\big)^\eta \le k \mathbb{E}_C\big[q_j(S[C]) - q_j(S)\big] + c_k u^{-k}.$$

For nonempty S, the nonnegative successive differences telescope to $q_d(S)$ because $q_0(S) = 0$. The power-sum inequality gives

$$q_d(S)^\eta \le d^{\eta-1} \sum_{j=0}^{d-1} \big(q_{j+1}(S) - q_j(S)\big)^\eta.$$

Summing the preceding estimate over j proves (7). For empty S, the left side of (7) is zero and the right side is nonnegative. □

The next form is the one used after conditioning the original nonuniform clause stream.

Lemma 2.3 (Robust potential drift)**.** *Let ν be a probability law on proper k-clauses on the u variables such that*

$$\nu(C) \ge B^{-1} U_{u,k}(C)$$

for every proper k-clause C, where $B \ge 1$. If $C \sim \nu$, then

$$\mathbf{1}_{\{S \ne \varnothing\}} q_d(S)^\eta \le k B d^{\eta-1} \mathbb{E}_\nu\big[P_d(S[C]) - P_d(S)\big] + c_k d^\eta u^{-k}. \tag{10}$$

Proof. The potential increment $P_d(S[C]) - P_d(S)$ is nonnegative for every clause C. Hence its expectation under ν is at least B^{-1} times its expectation under $U_{u,k}$. Apply Lemma 2.2. □

3 Clause omission

Fix $\rho < \infty$, let $1 \le L \le \rho n$, and put

$$T = H_n \wedge L.$$

For each $1 \le i \le L$, omit the clause at index i while retaining the original indices:

$$F_m^{(-i)} = \bigwedge_{\substack{1 \le j \le m \\ j \ne i}} C_j.$$

Let $T^{(-i)}$ be the first index at which this omitted-clause process is unsatisfiable, capped at L, and set

$$D_i = T^{(-i)} - T \ge 0.$$

The use of the original indices is important because the clause laws may depend on time.

Lemma 3.1 (Coordinate omission)**.** *The capped hitting time satisfies*

$$\operatorname{Var}(T) \le \sum_{i=1}^{L} \mathbb{E} D_i^2. \tag{11}$$

Proof. This is the coordinate-omission form of the Efron–Stein inequality; see Efron and Stein [1981] and Boucheron et al. [2005, Section 3.2]. For completeness, let

$$\mathcal{F}_i = \sigma(C_1, \dots, C_i), \qquad \Delta_i X = \mathbb{E}[X \mid \mathcal{F}_i] - \mathbb{E}[X \mid \mathcal{F}_{i-1}].$$

Because the clause coordinates are independent and $T^{(-i)}$ does not depend on C_i, one has $\Delta_i T^{(-i)} = 0$. Therefore $\Delta_i T = -\Delta_i D_i$. Conditional expectation is an orthogonal projection in L^2, so

$$\mathbb{E}(\Delta_i D_i)^2 \le \mathbb{E} D_i^2.$$

Summing the orthogonal martingale differences of T proves (11). □

It remains to prove a uniform bound on $\mathbb{E} D_i^2$. We do this after conditioning on the value of the omitted clause.

4 Root exposure and conditional comparability

Assume for now that $n \ge 2k$, so that $u = n - k \ge k$. Fix $i \le L$ and condition on an arbitrary value of C_i. Write R for its set of k variables, called the *roots*, and let $w \in \{0,1\}^R$ be the unique root assignment falsifying C_i.

For every $j \le L$, $j \ne i$, first reveal the exact subset of roots used by C_j and all literal signs on that subset. Classify the index as follows.

- If C_j meets no root, call it *ordinary* and leave all of its variables and signs hidden.
- If C_j meets exactly one root and its root literal is true under w, call it a *test clause* for that root and leave its $k - 1$ nonroot literals hidden.
- In every other case, reveal the entire clause.

Let $\mathcal{E}$ be the sigma-field generated by this exposure, including the clause-type schedule at future indices. Define

$$Z = \#\{j \le L : j \ne i,\ C_j \text{ meets } R\}, \tag{12}$$

$$V = \#\{j \le L : j \ne i,\ C_j \text{ meets at least two roots}\}, \tag{13}$$

$$d = \max\{1, Z\}. \tag{14}$$

Clauses counted by V are called *collisions*.

Because the clause coordinates are independent and the exposure conditions separately on each coordinate, the hidden parts remain conditionally independent across clause indices. Under the local criterion, their density bounds are precisely assumptions (i)–(ii). The following lemma verifies those assumptions from the simpler pointwise hypothesis used in Theorem 1.1.

Lemma 4.1 (Pointwise comparison implies conditional comparison)**.** *Under (1), the conditional law of each hidden ordinary clause has density between a/b and b/a relative to $U_{u,k}$. The conditional law of each hidden test part has density between a/b and b/a relative to $U_{u,k-1}$.*

Proof. For a fixed index j, write

$$f_j = \frac{\mathrm{d}\mu_{n,j}}{\mathrm{d}U_{n,k}}.$$

The pointwise assumption (1) gives $a \le f_j \le b$. Let E_j denote the root subset and root signs revealed at index j. Relative to the uniform conditional law of the hidden nonroot part, the conditional density under $\mu_{n,j}$ is

$$\frac{f_j(C_j)}{\mathbb{E}_{U_{n,k}}[f_j(C_j) \mid E_j]}.$$

Both numerator and denominator lie in $[a, b]$, so this ratio lies in $[a/b, b/a]$. Under the uniform law, conditioning on no roots leaves a uniform proper k-clause on the u nonroots, while conditioning on a specified single root and sign leaves a uniform proper $(k-1)$-clause on the nonroots. □

For $i \le m \le L$, let $S_m \subseteq \{0,1\}^u$ be the set of nonroot assignments z such that (w, z) satisfies $F_m^{(-i)}$. The sets S_m decrease with m. Define the filtrations

$$\mathcal{G}_m = \mathcal{E} \vee \sigma(\text{hidden ordinary-clause contents through index } m),$$
$$\mathcal{H}_m = \mathcal{G}_m \vee \sigma(\text{hidden test parts through index } m).$$

The set S_m is $\mathcal{G}_m$-measurable because every test clause is already satisfied by its root literal under w, regardless of the hidden nonroot part.

For $i \le m < L$, put

$$A_m = \{F_m^{(-i)} \text{ is satisfiable and } F_m^{(-i)} \wedge C_i \text{ is unsatisfiable}\}. \tag{15}$$

Then

$$A_m = \{T \le m < T^{(-i)}\}, \qquad D_i = \sum_{m=i}^{L-1} \mathbf{1}_{A_m}. \tag{16}$$

The event A_m is $\mathcal{H}_m$-measurable. On A_m, the set S_m is nonempty, and every solution of $F_m^{(-i)}$ has root assignment w.

Finally, put

$$p_m = q_d(S_m). \tag{17}$$

Then p_m is $\mathcal{G}_m$-measurable and nondecreasing in m.

5 Necessary root tests

A root $x \in R$ is called *available at time* m if no collision through time m, excluding index i, has its literal on x as its unique root literal true under w. A single collision excludes at most one root. Thus all k roots are available on $\{V = 0\}$ and at least $k - 1$ roots are available on $\{V = 1\}$.

Lemma 5.1 (Robust root-test bounds)**.** *For $i \le m < L$, conditional on $\mathcal{G}_m$,*

$$\mathbb{P}(A_m \mid \mathcal{G}_m) \le B_{\mathrm{t}}^{d} \mathbf{1}_{\{S_m \neq \varnothing\}} p_m^{k} \qquad \text{on } \{V = 0\}, \tag{18}$$

$$\mathbb{P}(A_m \mid \mathcal{G}_m) \le B_{\mathrm{t}}^{d} \mathbf{1}_{\{S_m \neq \varnothing\}} p_m^{k-1} \qquad \text{on } \{V = 1\}. \tag{19}$$

Proof. Fix an available root x. On A_m, the hidden nonroot parts of all test clauses for x through time m must kill S_m. Otherwise there is some $z \in S_m$ satisfying every one of those test parts. Flip only x in the assignment (w, z). Every clause of $F_m^{(-i)}$ remains satisfied:

- a clause with no root literal true under w already has a satisfied nonroot literal, which is unchanged;
- a clause with another root literal true under w retains that true literal;
- a clause whose unique true root literal is on x cannot be a collision, by availability, and is therefore a test clause whose nonroot part is satisfied by z.

The flipped assignment also satisfies C_i, contradicting A_m.

Suppose root x has r_x test clauses through time m. By the conditional test-density hypothesis, their joint density relative to r_x independent uniform proper $(k-1)$-clauses is at most $B_{\mathrm{t}}^{r_x}$. Hence their probability of killing S_m is at most

$$B_{\mathrm{t}}^{r_x} q_{r_x}(S_m) \le B_{\mathrm{t}}^{r_x} q_d(S_m) = B_{\mathrm{t}}^{r_x} p_m,$$

because $r_x \le Z \le d$. Test families belonging to distinct roots are conditionally independent. The total number of test clauses across any selected available roots is at most $Z \le d$. Multiplying the necessary events for all k available roots when $V = 0$, or for any $k - 1$ available roots when $V = 1$, proves (18) and (19). □

6 Occupation and remaining duration

The next estimate is the robust analogue of the critical occupation bound in OpenAI [2026b]. It uses the lower conditional-density bound for each ordinary clause.

Lemma 6.1 (Conditional occupation estimate)**.** *There is a constant $K_0 = K_0(B_{\mathrm{o}}, k, \rho) < \infty$ such that, for every $i \le m < L$,*

$$\mathbb{E}\left[\sum_{s=m}^{L-1} \mathbf{1}_{\{S_s \neq \varnothing\}} p_s^{\eta} \,\middle|\, \mathcal{H}_m\right] \le K_0 d^{\eta}. \tag{20}$$

Proof. Suppose $s+1$ is ordinary, a fact known under $\mathcal{E}$. Then $S_{s+1} = S_s[C_{s+1}]$, and the hidden contents of C_{s+1} remain fresh given $\mathcal{H}_s$. By the ordinary-clause density hypothesis, their conditional law is bounded below by $B_{\mathrm{o}}^{-1} U_{u,k}$. Apply Lemma 2.3 conditionally to obtain

$$\begin{aligned} \mathbf{1}_{\{S_s \neq \varnothing\}} p_s^{\eta} \le{}& k B_{\mathrm{o}}\, d^{\eta-1} \mathbb{E}[P_d(S_{s+1}) - P_d(S_s) \mid \mathcal{H}_s] \\ &+ c_k d^{\eta} u^{-k}. \end{aligned} \tag{21}$$

Condition on $\mathcal{H}_m$ and sum over ordinary steps. Every potential increment is nonnegative, including increments at nonordinary steps. Thus, pathwise,

$$\sum_{\substack{m\le s<L\\ s+1 \text{ ordinary}}} \big(P_d(S_{s+1}) - P_d(S_s)\big) \le P_d(S_L) - P_d(S_m) \le d.$$

The ordinary-step contribution to (20) is therefore at most

$$kB_{\mathrm{o}}\, d^\eta + c_k L d^\eta u^{-k}.$$

There are at most Z remaining steps, because every nonordinary index strictly after i meets at least one root. Each remaining summand is at most one. Since $Z \le d \le d^\eta$, $L \le \rho n$, and $u = n - k \ge n/2$, we obtain

$$\mathbb{E}\left[\sum_{s=m}^{L-1} \mathbf{1}_{\{S_s\neq\varnothing\}} p_s^\eta \,\middle|\, \mathcal{H}_m\right] \le \left(kB_{\mathrm{o}} + 1 + c_k\rho 2^k\right) d^\eta.$$

This proves the assertion with, for example, $K_0 = kB_{\mathrm{o}} + 1 + c_k\rho 2^k$. □

Corollary 6.2 (Remaining duration)**.** *For $i \le m < L$,*

$$\mathbb{E}\left[\sum_{s=m}^{L-1} \mathbf{1}_{A_s} \,\middle|\, \mathcal{H}_m\right] \le W_m, \tag{22}$$

where

$$W_m = \begin{cases} \min\{L, K_0 d^\eta p_m^{-\eta}\}, & p_m > 0,\\ L, & p_m = 0. \end{cases} \tag{23}$$

In particular, W_m is $\mathcal{G}_m$-measurable.

Proof. For $s \ge m$, one has $p_s \ge p_m$, and A_s implies $S_s \neq \varnothing$. If $p_m > 0$, then pathwise

$$\sum_{s=m}^{L-1} \mathbf{1}_{A_s} \le p_m^{-\eta} \sum_{s=m}^{L-1} \mathbf{1}_{\{S_s\neq\varnothing\}} p_s^\eta.$$

Apply Lemma 6.1. The count is always at most L, which also gives the case $p_m = 0$. Measurability follows from that of d and p_m. □

7 Squared deletion delays

We now combine the root-test probability with the remaining-duration bound. The two estimates use compatible filtrations: A_m is known under $\mathcal{H}_m$, while the upper bound W_m is already measurable under $\mathcal{G}_m$ and therefore does not reveal the hidden root tests.

Proposition 7.1 (Conditional squared-delay bound)**.** *For every $1 \le i \le L$,*

$$\begin{aligned}\mathbb{E}[D_i^2 \mid \mathcal{E}] \le{}& 2K_0^2 B_{\mathrm{t}}^d\, d^{2\eta} \mathbf{1}_{\{V=0\}}\\ &+ 2K_0 L B_{\mathrm{t}}^d\, d^\eta \mathbf{1}_{\{V=1\}} + L^2 \mathbf{1}_{\{V\ge 2\}}.\end{aligned} \tag{24}$$

Proof. For $i = L$, the delay is zero. Otherwise, (16) implies

$$D_i^2 \le 2 \sum_{m=i}^{L-1} \mathbf{1}_{A_m} \sum_{s=m}^{L-1} \mathbf{1}_{A_s}.$$

Condition first on $\mathcal{H}_m$, where A_m is known, and apply Corollary 6.2. Then condition on $\mathcal{G}_m$, where W_m is known. Conditioning the resulting estimates down to $\mathcal{E}$ gives

$$\mathbb{E}[D_i^2 \mid \mathcal{E}] \le 2 \sum_{m=i}^{L-1} \mathbb{E}[W_m \mathbb{P}(A_m \mid \mathcal{G}_m) \mid \mathcal{E}]. \tag{25}$$

On $\{V = 0\}$, Lemma 5.1 and (23) bound the product inside the expectation by

$$K_0 B_{\mathrm{t}}^d \, d^\eta \mathbf{1}_{\{S_m \neq \varnothing\}} p_m^{k-\eta}.$$

For every $k \ge 3$,

$$k - \eta \ge \eta, \tag{26}$$

because $k \ge 2k/(k-1)$. Since $0 \le p_m \le 1$, the preceding display is at most

$$K_0 B_{\mathrm{t}}^d \, d^\eta \mathbf{1}_{\{S_m \neq \varnothing\}} p_m^{\eta}.$$

Lemma 6.1, applied from time i and then conditioned down to $\mathcal{E}$, bounds the sum in (25) by $2K_0^2 B_{\mathrm{t}}^d \, d^{2\eta}$.

On $\{V = 1\}$, the corresponding product is at most

$$K_0 B_{\mathrm{t}}^d \, d^\eta p_m^{k-1-\eta} \le K_0 B_{\mathrm{t}}^d \, d^\eta,$$

because $k - 1 - \eta \ge 0$ for $k \ge 3$. There are at most L terms. If $p_m = 0$, the conditional pivotality probability is zero by Lemma 5.1, so no division by zero occurs. Finally, on $\{V \ge 2\}$ use $D_i \le L$. □

It remains to average the factors in (24). The clause coordinates need not have the same law, but the local incidence hypotheses give uniform incidence probabilities.

Lemma 7.2 (Incidence and collision moments)**.** *Uniformly in $n \ge 2k$, $i \le L$, and the conditioned value of C_i,*

$$\mathbb{E}[B_{\mathrm{t}}^d \, d^{2\eta}] = O(1), \tag{27}$$

$$\mathbb{E}[B_{\mathrm{t}}^d \, d^\eta \mathbf{1}_{\{V \ge 1\}}] = O(n^{-1}), \tag{28}$$

$$\mathbb{P}(V \ge 2) = O(n^{-2}). \tag{29}$$

The implicit constants depend only on $k, \rho, B_{\mathrm{t}}, \gamma_1, \gamma_2$.

Proof. For $j \neq i$, let I_j be the indicator that C_j meets at least one root, and let J_j be the indicator that it meets at least two roots. The local incidence assumptions give

$$\mathbb{P}(I_j = 1) \le \frac{\gamma_1}{n}, \qquad \mathbb{P}(J_j = 1) \le \frac{\gamma_2}{n(n-1)}. \tag{30}$$

The indicators at distinct indices are independent, although their probabilities may differ. Consequently, for every fixed $t \geq 0$,

$$\mathbb{E}e^{tZ} = \prod_{\substack{j \leq L \\ j \neq i}} \Big(1 + (e^t - 1)\mathbb{P}(I_j = 1)\Big) \leq \exp\big((e^t - 1)\gamma_1 \rho\big).$$

Choosing $t > \log B_{\mathrm{t}}$ and absorbing the polynomial factor $d^{2\eta}$ into a slightly larger exponential proves (27).

Conditional on $J_j = 1$, the other clause coordinates retain their original independent laws, and Z is one plus the incidence count from those other coordinates. The same exponential-moment estimate therefore gives

$$\mathbb{E}[B_{\mathrm{t}}^d \, d^\eta \mid J_j = 1] = O(1)$$

uniformly in j. Since $\mathbf{1}_{\{V \geq 1\}} \leq \sum_{j \neq i} J_j$,

$$\mathbb{E}[B_{\mathrm{t}}^d \, d^\eta \mathbf{1}_{\{V \geq 1\}}] \leq O(1) \sum_{j \neq i} \mathbb{P}(J_j = 1) = O(n^{-1}),$$

which proves (28). Finally, independence across indices gives

$$\mathbb{P}(V \geq 2) \leq \mathbb{E}\binom{V}{2} = \sum_{j < \ell} \mathbb{P}(J_j = 1)\mathbb{P}(J_\ell = 1) = O(n^{-2}).$$

□

Under the pointwise comparison hypothesis, the concrete bounds (2) follow because, under $U_{n,k}$, a specified root is included with probability k/n and a specified root pair with probability $k(k-1)/[n(n-1)]$; union bounds over roots and root pairs and the upper density factor b give (30).

Proposition 7.3 (Uniform deletion-delay moment)**.** *For every fixed $\rho < \infty$,*

$$\sup_{\substack{n \geq 2k,\, 1 \leq L \leq \rho n \\ 1 \leq i \leq L}} \mathbb{E}D_i^2 < \infty,$$

where the supremum is uniform over all clause-law arrays satisfying the local hypotheses of Theorem 1.2.

Proof. Average (24). The first term is $O(1)$ by (27). The second is

$$O(L)\,\mathbb{E}[B_{\mathrm{t}}^d \, d^\eta \mathbf{1}_{\{V \geq 1\}}] = O(Ln^{-1}) = O(1),$$

and the third is

$$L^2 \mathbb{P}(V \geq 2) = O(L^2 n^{-2}) = O(1).$$

The bounds were uniform in the conditioned value of C_i, so that conditioning may be removed. □

Combining Proposition 7.3 with Lemma 3.1 proves

$$\mathrm{Var}(H_n \wedge L) = O(n) \qquad (n \geq 2k,\ L \leq \rho n), \tag{31}$$

with the dependence asserted in Theorem 1.2. For the finitely many sizes $k \leq n < 2k$, the deterministic bound $H_n \wedge L \leq L \leq \rho n$ gives the same $O(n)$ conclusion after increasing the constant. This proves the clipped part of the local criterion.

8 Removing the cap and proving the consequences

Let δ be the uniform assignment-falsification constant from assumption (iv), and put

$$r = 1 - \delta < 1.$$

For every fixed assignment, independence across clause times gives

$$\mathbb{P}(\text{a fixed assignment satisfies } F_m) \le r^m.$$

A union bound over the 2^n assignments yields

$$\mathbb{P}(H_n > m) \le 2^n r^m. \tag{32}$$

In particular, H_n is almost surely finite and has a finite second moment.

Choose

$$L_n = \left\lceil \frac{\log 2 + 1}{\delta}\, n \right\rceil, \qquad T_n = H_n \wedge L_n. \tag{33}$$

Since $r \le e^{-\delta}$,

$$2^n r^{L_n} \le e^{-n}.$$

The integer-valued tail-sum identity gives

$$\begin{aligned}
\mathbb{E}(H_n - T_n)^2 &= \sum_{j=0}^{\infty} (2j+1)\mathbb{P}(H_n > L_n + j) \\
&\le 2^n r^{L_n} \sum_{j=0}^{\infty} (2j+1) r^j \\
&= 2^n r^{L_n} \frac{1+r}{(1-r)^2} = O_\delta(e^{-n}).
\end{aligned} \tag{34}$$

The cap in (33) is a fixed linear multiple of n. Thus (31) gives $\mathrm{Var}(T_n) = O(n)$ for $n \ge 2k$, with dependence on the local parameters. Consequently,

$$\mathrm{Var}(H_n) \le 2\,\mathrm{Var}(T_n) + 2\,\mathrm{Var}(H_n - T_n) \le 2\,\mathrm{Var}(T_n) + 2\mathbb{E}(H_n - T_n)^2 = O(n).$$

For the finitely many sizes $k \le n < 2k$, equation (32) directly bounds $\mathbb{E}H_n^2$ by a constant depending only on δ and k. Enlarging the constants proves the uncapped assertion of Theorem 1.2. The parameter choices in (2) then prove Theorem 1.1.

Proof of Corollary 1.3. Let $\mu_n = \mathbb{E}H_n$, and let K be the constant in Theorem 1.1. Put

$$t_n = \sqrt{\frac{2Kn}{\varepsilon}}.$$

Chebyshev's inequality gives

$$\mathbb{P}(|H_n - \mu_n| \ge t_n) \le \frac{\varepsilon}{2}.$$

At $m = \lceil \mu_n + t_n \rceil$, this implies $s_n(m) \le \varepsilon/2$, so $r_n(\varepsilon) \le \lceil \mu_n + t_n \rceil$. For every nonnegative integer $m \le \mu_n - t_n$, one has $s_n(m) \ge 1 - \varepsilon/2 > 1 - \varepsilon$, so $r_n(1-\varepsilon) > \mu_n - t_n$ unless the latter quantity is negative, in which case the trivial bound $r_n(1-\varepsilon) \ge 0$ suffices. Hence

$$r_n(\varepsilon) - r_n(1-\varepsilon) \le 2t_n + 2.$$

□

9 Scope and further directions

The local criterion separates the four inputs that the proof actually uses. A global upper density bound is unnecessary if root-test densities and root incidences are controlled directly. Thus variable-weighted or otherwise nonuniform models may fall within the theorem even when their full clause law is not bounded above by a constant multiple of the uniform law. The ordinary-clause lower density controls the potential drift, the test-clause upper density controls pivotality, the incidence conditions control the robustness factors and collisions, and the assignment-falsification bound is needed only to remove the cap.

Under the simpler pointwise hypothesis, broad nonuniformity is allowed. For example, at time j the unordered variable set may have probability between fixed multiples of $\binom{n}{k}^{-1}$ and, conditional on that set, each sign pattern may have probability between fixed multiples of 2^{-k}. No convergence, periodicity, or stationarity of these laws is required.

Independence between clause times is used in the coordinate-omission argument, the product structure of hidden root tests, the incidence moments, and the uncapping tail. Adaptive clause laws or an unconditioned common random environment are therefore not covered as stated.

A matching linear lower variance bound remains the principal open question. The mixture decomposition supplied by a pointwise lower density does not by itself prevent the residual stream from determining the hitting time. The present proof also does not automatically yield subgaussian tails. The Boucheron–Bousquet–Lugosi–Massart moment method would require an almost sure or sufficiently high-moment control of the full deletion variance proxy, whereas here each D_i can be of order n and only $\mathbb{E}D_i^2 = O(1)$ is established. Either strengthening would require a new anti-concentration or high-moment ingredient rather than a formal reuse of the variance proof.

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